DP Biology – C3.1 (HL extension)

How plants respond and grow

- What do nerves and hormones each do to coordinate body systems?
- How does feedback help to control body systems?

HL statements C3.1.17–23 (plants). Textbook pp. 509–517.



Image: an orchid on a windowsill, leaning towards the light. Tangopaso, CC BY-SA 3.0.

Objectives for each class

Class	Statements	By the end of the class, you can...
1	C3.1.17–18	Observe and measure tropic responses, and explain why shoots grow towards light
2	C3.1.19–21	Describe phytohormones, and explain how auxin is moved and how it makes cells elongate
3	C3.1.22–23	Explain how auxin and cytokinin balance root and shoot growth, and how ethylene drives ripening

Plants respond without nerves

C3.1.17



A houseplant curving towards a window. Tangopaso, CC BY-SA 3.0.

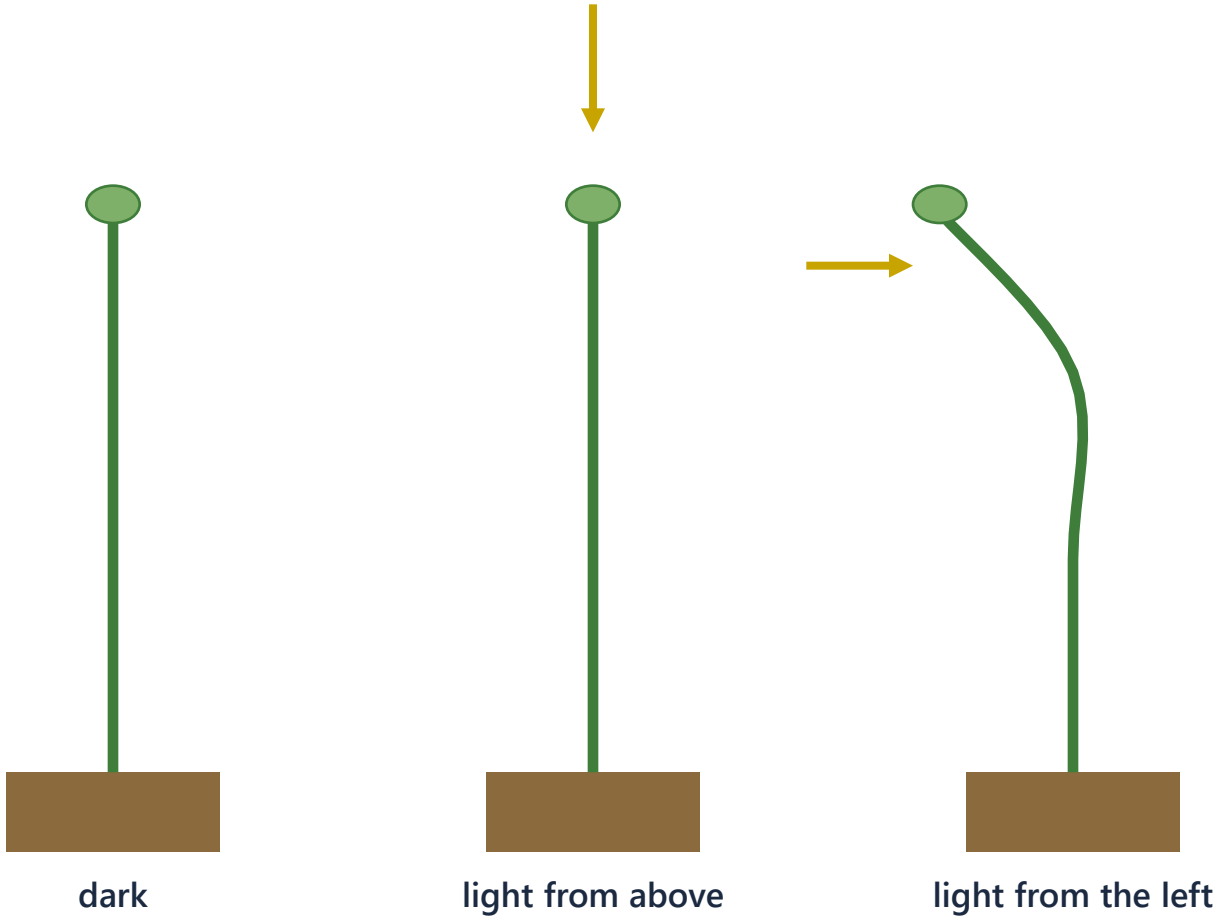
Plants cannot move away, so they respond by **growing**. A **tropism** is growth in a direction set by a stimulus that comes from one side.

The signals are chemicals: **phytohormones**.

Examples

Shoots grow towards light (phototropism). Roots grow down with gravity (gravitropism). Tendrils curl round supports.

Observing tropic responses



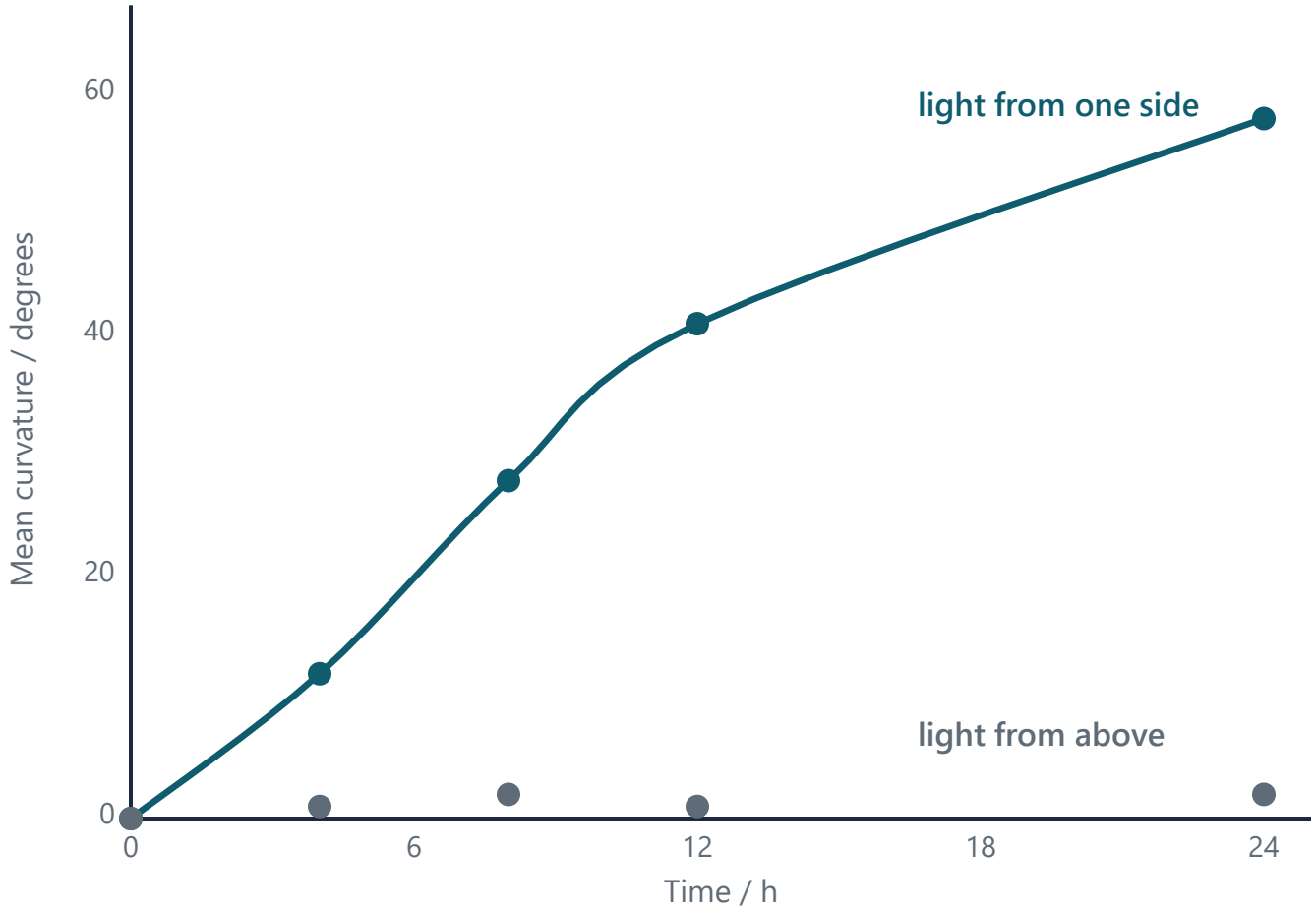
Drawn seedlings after a day.

Grow seedlings in the dark, then light them from one side.

Record drawings (**qualitative**) and the angle of curvature (**quantitative**). The dark and lit-from-above groups are **controls**.

Working with data: curvature over time

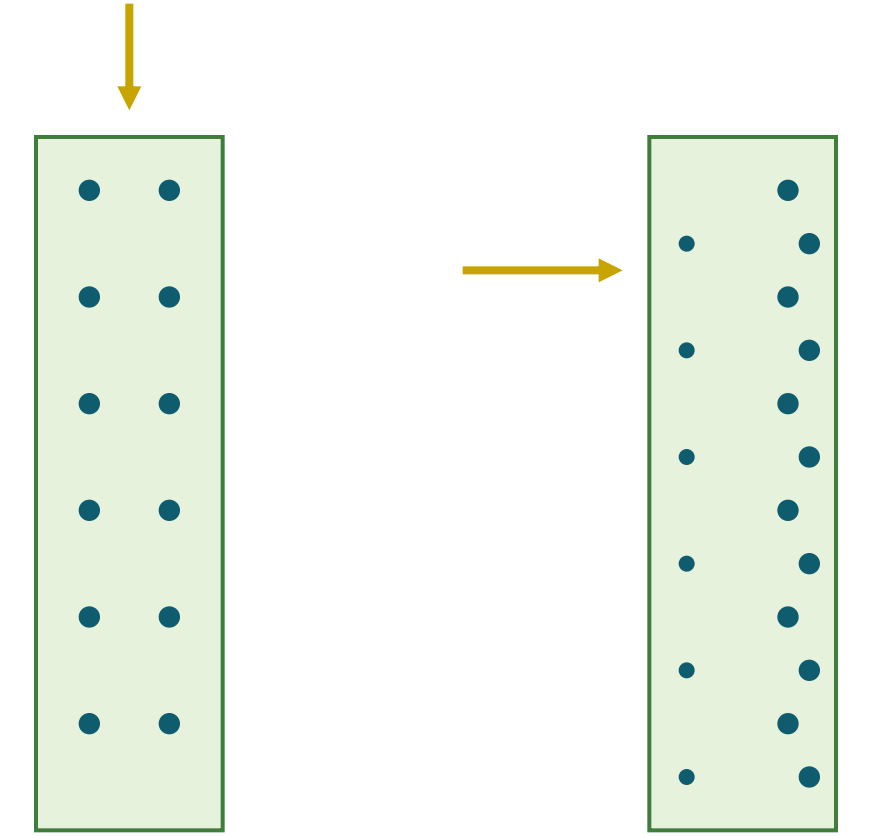
C3.1.17



Invented example: mean of 20 seedlings per group.

1. Describe the trend for light from one side.
2. Calculate the mean rate of bending in the first 12 h.
3. Explain why the lit-from-above group is needed.
4. Suggest why bending slows after 12 h.

Positive phototropism



Light from above

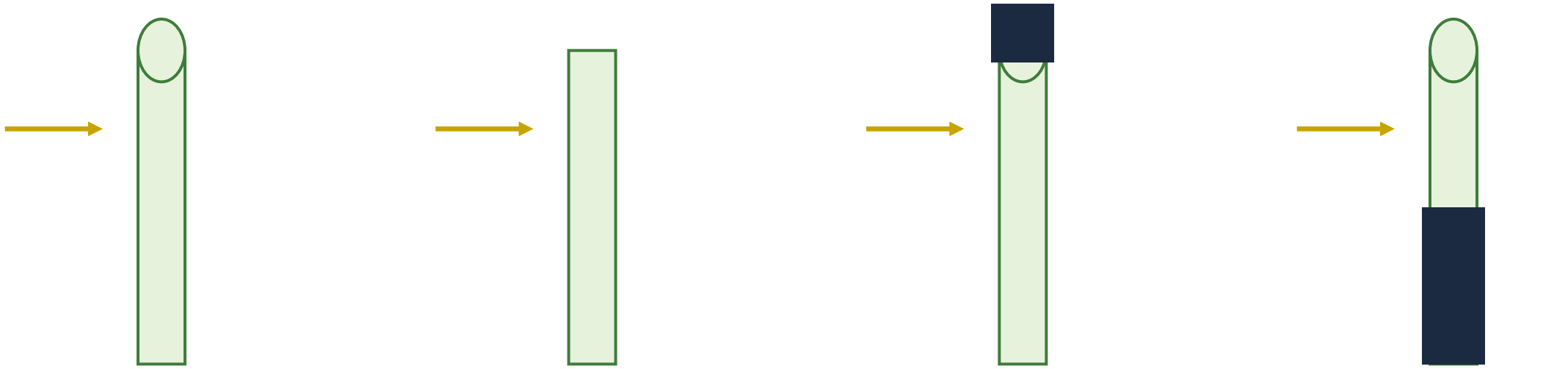
Light from the left

Dots: auxin. Drawn shoot tips, before bending.

Light from one side is detected near the tip. **Auxin** is moved to the **shaded** side.

Shaded cells elongate more, so the shoot bends **towards** the light. Leaves then catch more light.

Where is the light detected?



Tip intact
Bends towards light

Tip removed
No bending

Tip covered
No bending

Base covered
Bends towards light

Conclusion

The tip detects light, but the bending happens lower down: a signal must travel.

Practice: predict the growth

C3.1.17–18

Spun on a turntable

Seedling rotated slowly in light from one side.

Foil cap on tip

Tip covered, light from the right.

Mica in the shaded side

A barrier blocks auxin moving down the shaded side.

For each one

Predict whether the shoot bends, and which way. Explain using auxin.

Check-in: class 1

C3.1.17–18

1. Define a tropism and give two examples.
2. Describe how to measure a tropic response.
3. Explain why controls are needed in the practical.
4. Explain how a shoot bends towards light.
5. State where light is detected in a shoot.

How to use this

Answer without your notes, then check against slides 3–8.

Phytohormones

C3.1.19

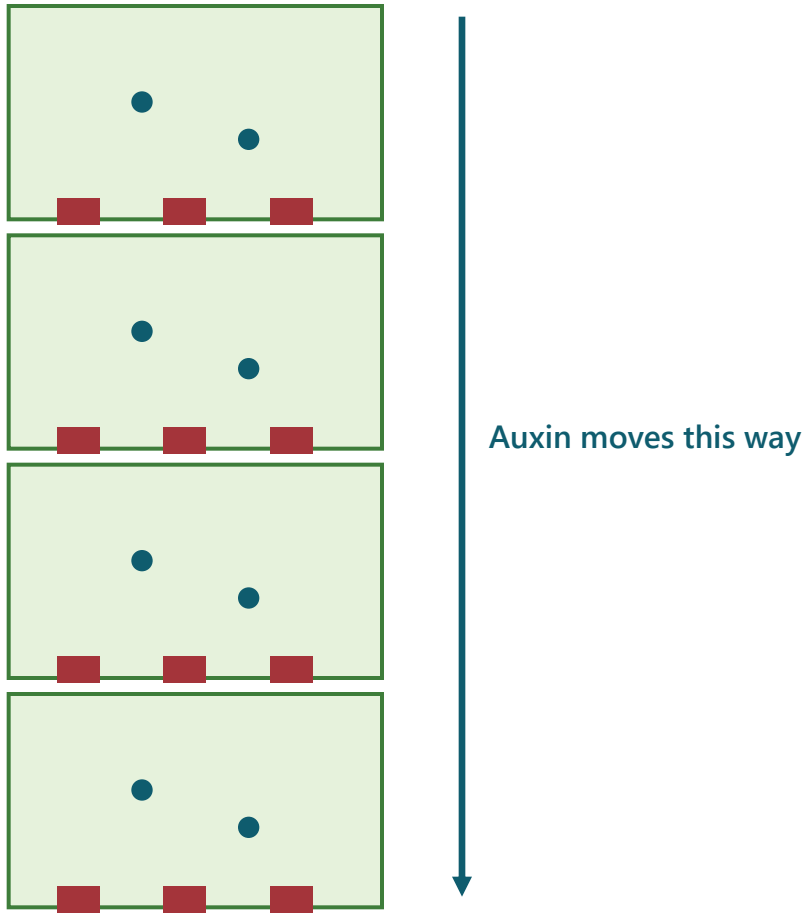
Phytohormone	Example of its effects
Auxin	Cell elongation; tropisms; apical dominance; root formation
Cytokinin	Cell division; growth of side buds
Gibberellin	Stem elongation; seed germination
Ethylene	Fruit ripening; leaf fall
Abscisic acid	Stomata close in drought; seed dormancy

Key idea

Plants use many chemical signals, often together, and each has several effects.

Auxin efflux carriers

C3.1.20



Red: efflux carriers, all on the same side of each cell. Dots: auxin.

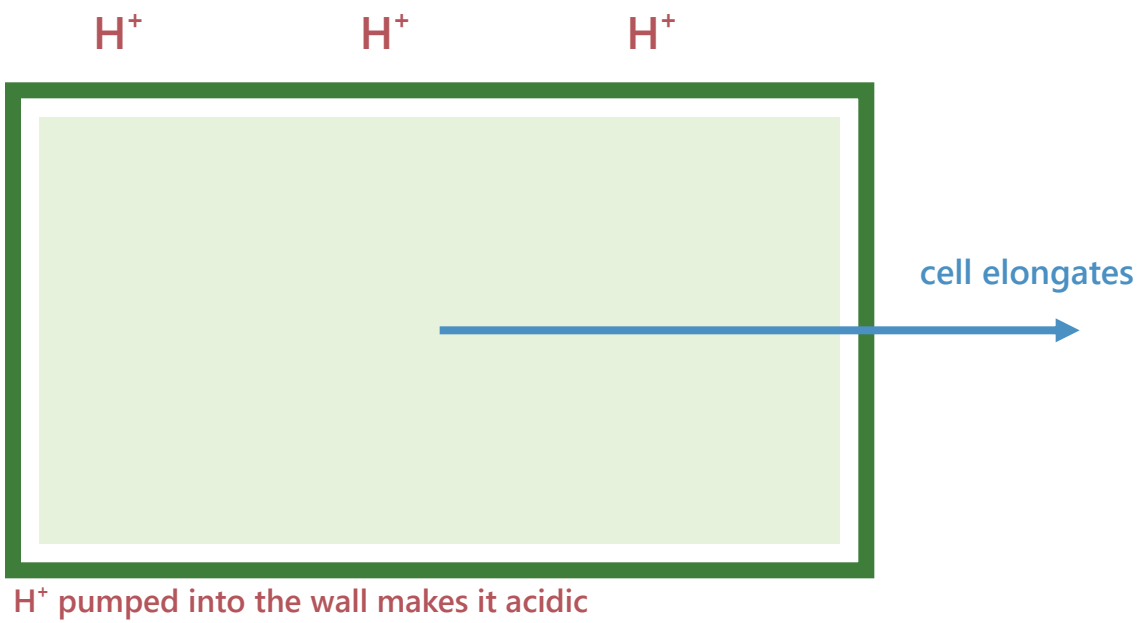
Auxin enters cells easily but can only leave through **efflux carriers** (PIN proteins), using energy.

When every cell puts its carriers on the same side, auxin is passed along in **one direction** and builds up where it is needed: a **gradient**.

How auxin makes cells grow

C3.1.21

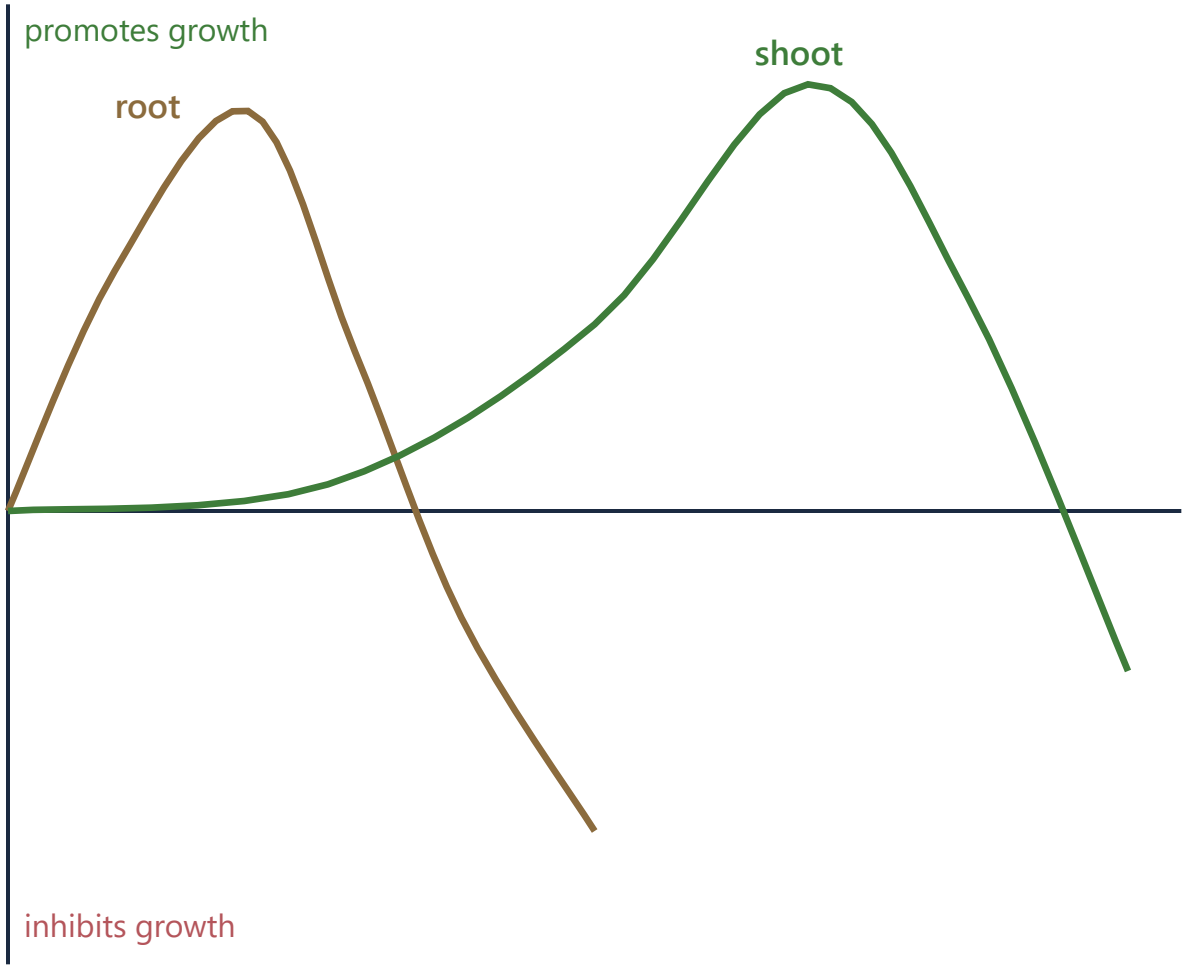
Auxin binds receptors → Proton pumps move H^+ into the wall → The wall becomes acidic → **Expansins loosen cellulose links** → Turgor stretches the cell



The **acid growth** idea: low pH in the wall lets it stretch, and water entering by osmosis pushes the cell longer.

Same hormone, different response

C3.1.21



Drawn sketch of the classic response curves.

Roots are far more **sensitive** to auxin than shoots.

A concentration that speeds up shoot growth can **slow** root growth.

Working with data: auxin and curvature

C3.1.19–21

Curvature of oat shoots after agar blocks with auxin were placed on one side of the cut tip. These data are invented for practice; the pattern matches classic experiments.

Auxin / mg dm^{-3}	Curvature / degrees
0	0
0.05	6
0.1	12
0.2	19
0.4	20

1. Describe the relationship.
2. Explain why shoots bend away from the block.
3. Explain why the block with no auxin is included.
4. Suggest why the curve levels off.

Practice: explain the gradient

C3.1.20–21

PIN drug

A drug stops efflux carriers working.

Neutral wall

A buffer keeps the cell wall at pH 7.

Root on its side

A root is laid flat; auxin collects underneath.

For each one

Predict the effect on growth or bending, and explain.

Check-in: class 2

C3.1.19–21

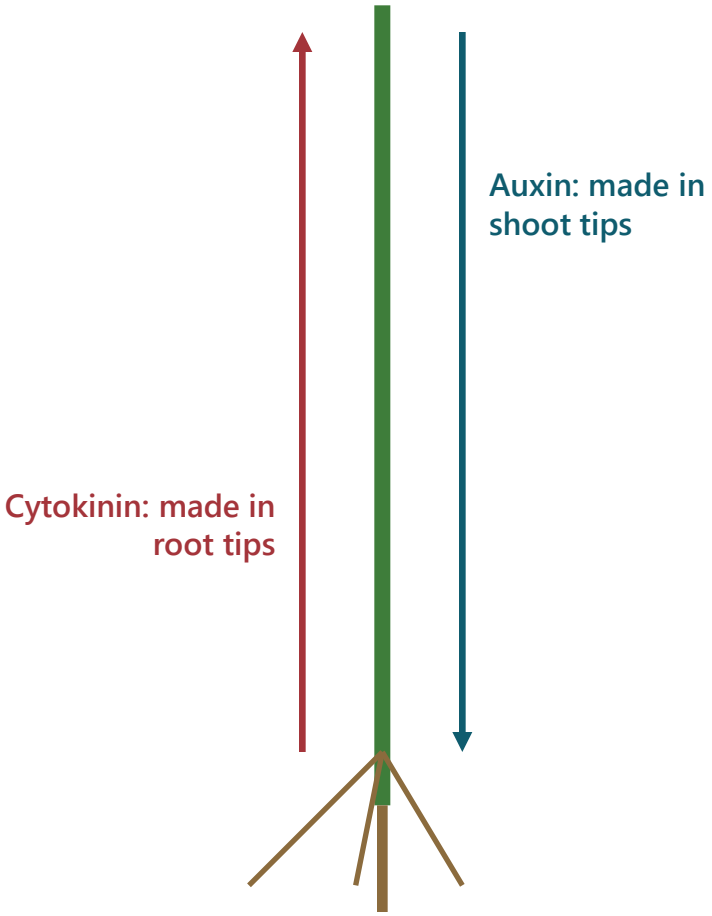
1. Give three phytohormones and one effect of each.
2. Explain how efflux carriers set up an auxin gradient.
3. Outline how auxin makes a cell elongate.
4. Explain why roots and shoots respond differently to auxin.
5. State the role of expansins.

How to use this

Answer without your notes, then check against slides 10–15.

Auxin and cytokinin work together

C3.1.22

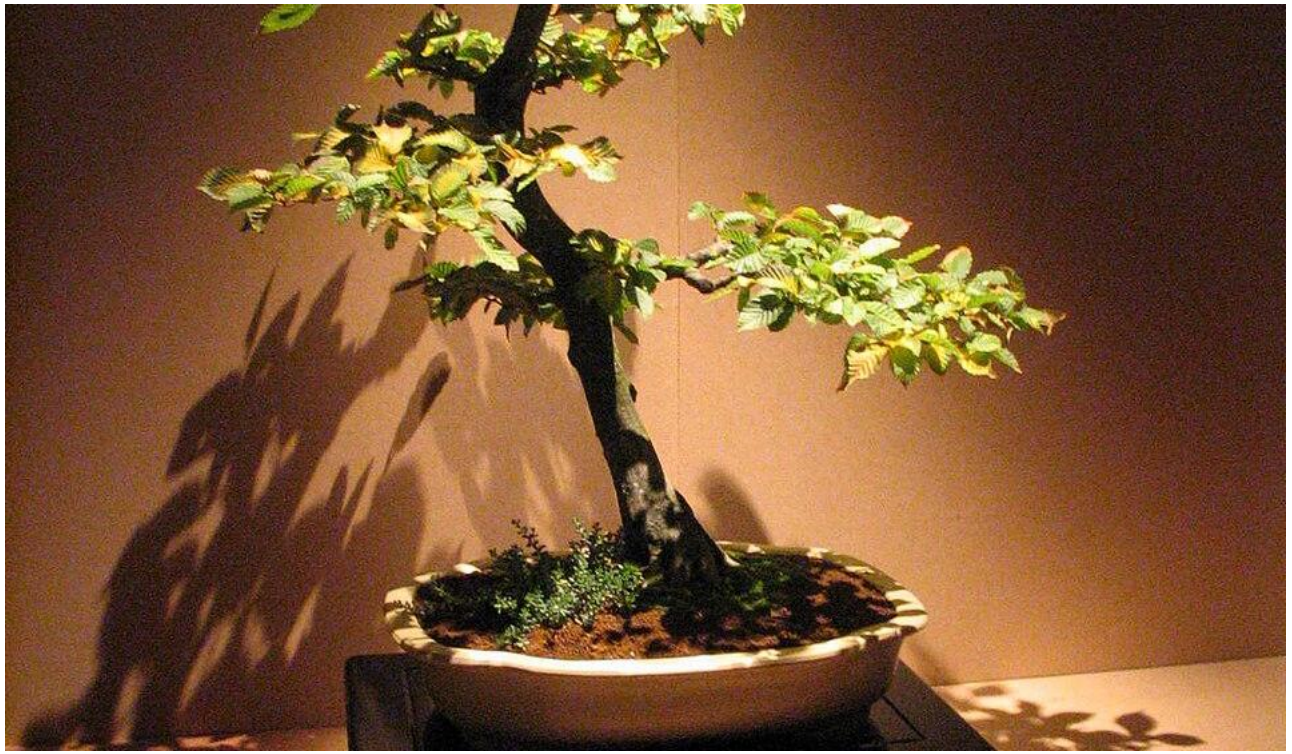


Auxin travels **down** from the shoot tips; cytokinin travels **up** from the root tips.

A big root system sends up more cytokinin, supporting more shoot growth. A big shoot sends down more auxin, supporting more roots. So roots and shoots stay **in proportion**.

Apical dominance and pruning

C3.1.22



Bonsai: pruning shapes growth. Dake, CC BY-SA 2.5.

Auxin from the main shoot tip stops side buds growing: **apical dominance**. Cytokinin from the roots promotes side buds.

Cut off the tip and auxin falls: side buds grow and the plant becomes **bushy**.

Think

A gardener wants a tall, single-stemmed sunflower. Should they pinch out the tip? Explain.

Practice: hormones in tissue culture

C3.1.22

Plant tissue grown on agar with different hormone ratios. Typical results.

Auxin : cytokinin	What grows
High : low	Roots
Equal	Undifferentiated callus
Low : high	Shoots

1. Which ratio gives roots?
2. Suggest a two-step plan to grow a whole plant from a callus.
3. Link the table to root and shoot growth in a whole plant.

Ethylene and ripening

C3.1.23



Tomatoes ripening on the plant. Naposia, CC BY-SA 3.0.

Ripening: starch turns to sugar, flesh softens, colour and scent change.

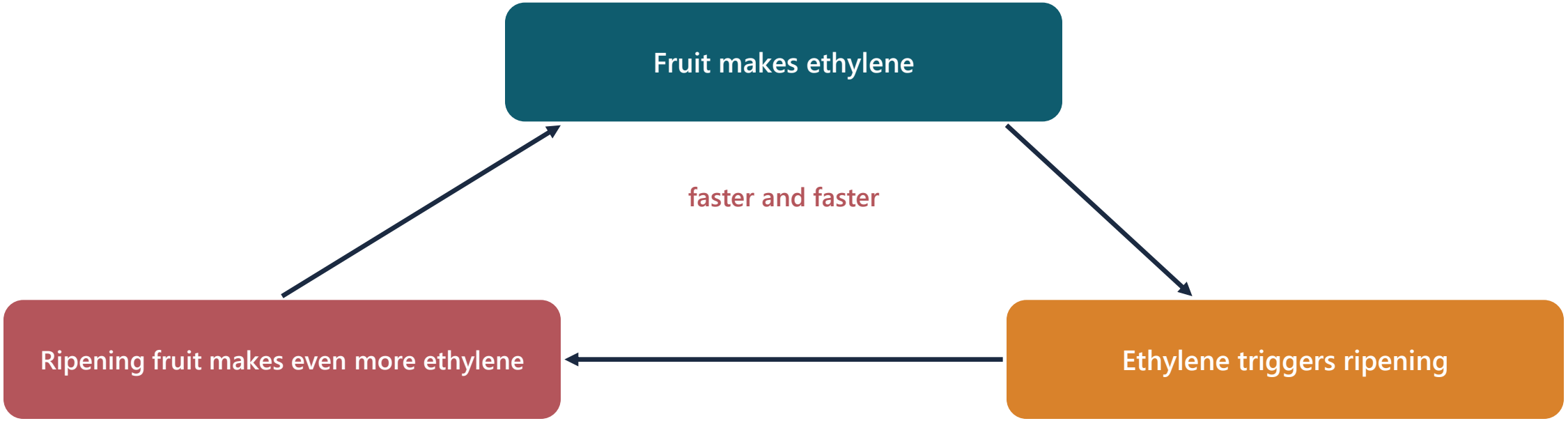
Ethylene, a gas, triggers ripening. It spreads to nearby fruit, so they ripen too.

Key idea

Ripening makes more ethylene, and more ethylene speeds ripening: positive feedback.

A self-reinforcing loop

C3.1.23



Why useful

All the fruits ripen quickly and together, so animals find a whole crop and disperse many seeds.

Ethylene in the food industry

C3.1.23

Bananas

Picked green, shipped cold, then ripened with ethylene.

Ethylene absorbers

Packets in packaging slow ripening.

One bad apple

A rotting apple releases ethylene and ripens the rest.

Think

Why are bananas shipped cold and in well-ventilated containers?

Working with data: ripening in a bag

C3.1.23

Days for green tomatoes to turn red in sealed bags. The values are invented for practice; the pattern is realistic.

Treatment	Days to ripen
Tomatoes alone	9
With a ripe banana	5
With a ripe banana and an ethylene absorber	9
Open tray, no bag	11

1. Explain the effect of the banana.
2. Explain the result with the absorber.
3. Explain why the open tray was slowest.
4. Name the type of feedback involved.

Check-in: class 3

C3.1.22–23

1. State where auxin and cytokinin are made.
2. Explain how the two keep root and shoot growth in balance.
3. Explain why removing a shoot tip makes a plant bushy.
4. Explain how ethylene coordinates ripening.
5. Explain why the loop in ripening is positive feedback.

How to use this

Answer without your notes, then check against slides 17–23.

Common mistakes to avoid

Not this	But this
Auxin moves towards the light.	Auxin is moved to the shaded side.
Light makes shoot cells grow.	Auxin on the shaded side makes cells elongate.
Auxin always speeds up growth.	High auxin slows root growth.
Ethylene is a liquid hormone.	Ethylene is a gas, so it spreads between fruits.

Summary: plant responses

C3.1.17–23

	Key idea
C3.1.17–18	Measure curvature; shoots bend towards one-sided light
C3.1.19	Phytohormones: auxin, cytokinin, gibberellin, ethylene, abscisic acid
C3.1.20	PIN carriers pass auxin one way, building a gradient
C3.1.21	Auxin: H^+ into the wall, expansins, cell elongates
C3.1.22	Auxin from shoots, cytokinin from roots: growth in proportion
C3.1.23	Ethylene and ripening reinforce each other

Check that you can...

1. **Describe** how to measure a tropic response
2. **Explain** positive phototropism using auxin
3. **Explain** how efflux carriers create auxin gradients
4. **Explain** how auxin causes cell elongation
5. **Outline** how auxin and cytokinin interact
6. **Explain** the role of ethylene and positive feedback in ripening

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. Define positive phototropism.
2. Explain how auxin causes cells to elongate.
3. Explain why fruits on a plant ripen at the same time.

Image credits

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Bonsai IMG 6426.jpg – Dake – CC BY-SA 2.5

Ripening tomatoes.jpg – Naposia – CC BY-SA 3.0

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